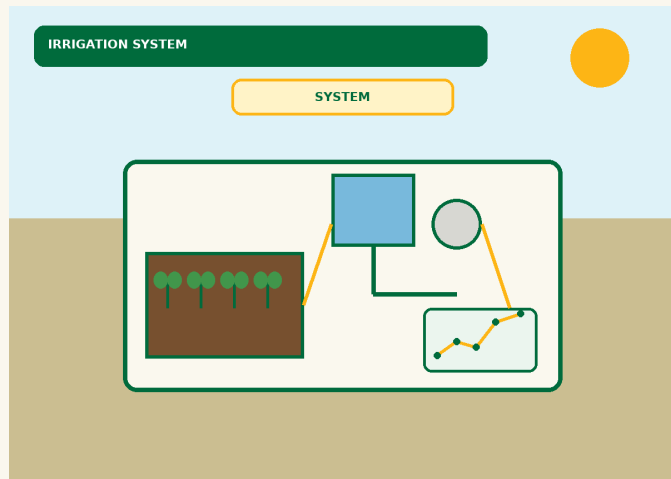




SYSTEM

IRRIGATION SYSTEM

DESIGN • INSTRUMENT • MODEL • EVALUATE • OPTIMIZE

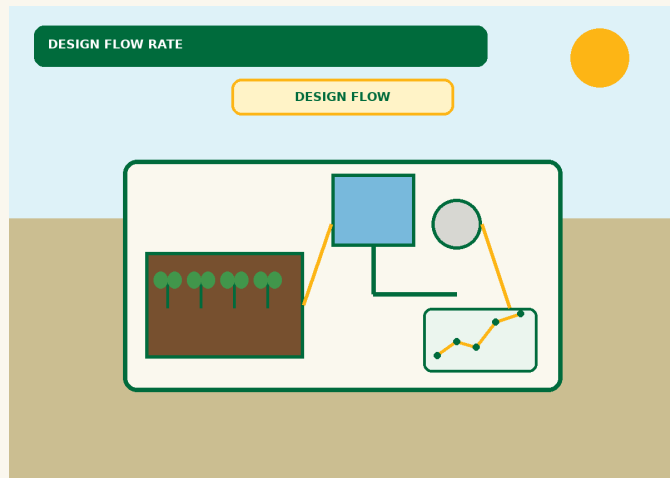
TECHNICAL CONCEPT

An irrigation system stores, conveys, controls, distributes, and applies water to meet defined crop and operational requirements.

ANALYSIS / DESIGN TASK

Draw a hydraulic system schematic.

DEFINE → MEASURE → MODEL → EVALUATE → DECIDE



DESIGN FLOW

DESIGN FLOW RATE

DESIGN • INSTRUMENT • MODEL • EVALUATE • OPTIMIZE

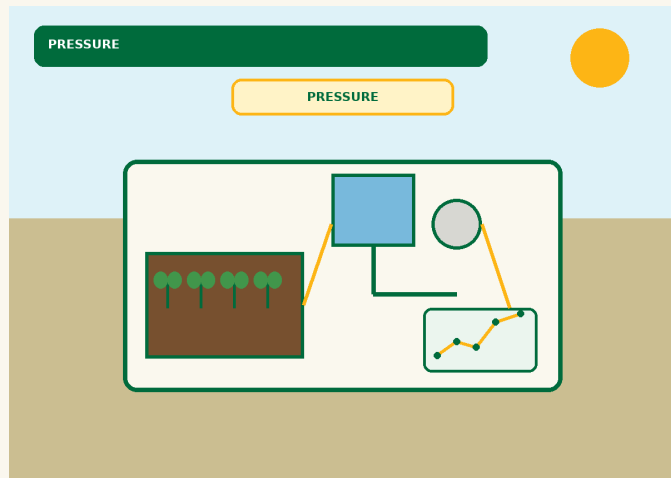
TECHNICAL CONCEPT

Design flow rate is the water discharge the system must deliver under specified operating conditions.

ANALYSIS / DESIGN TASK

Calculate required flow from emitter number and discharge.

DEFINE → MEASURE → MODEL → EVALUATE → DECIDE



PRESSURE

PRESSURE

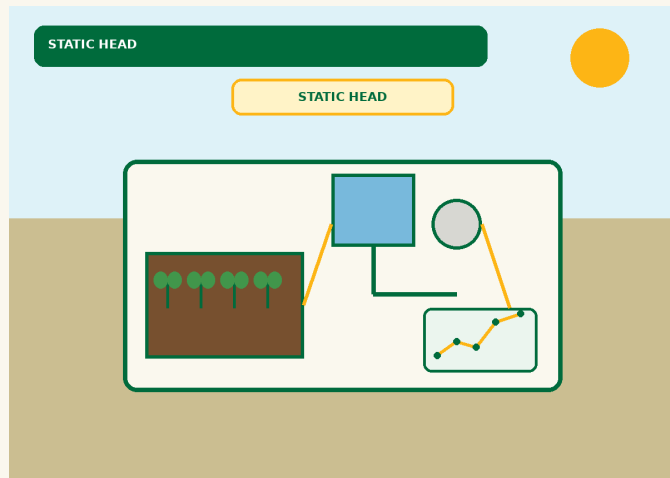
DESIGN • INSTRUMENT • MODEL • EVALUATE • OPTIMIZE

TECHNICAL CONCEPT

Pressure is force per unit area and provides hydraulic energy for water movement through pressurized irrigation systems.

ANALYSIS / DESIGN TASK

Interpret a pressure-gauge reading.



STATIC HEAD

STATIC HEAD

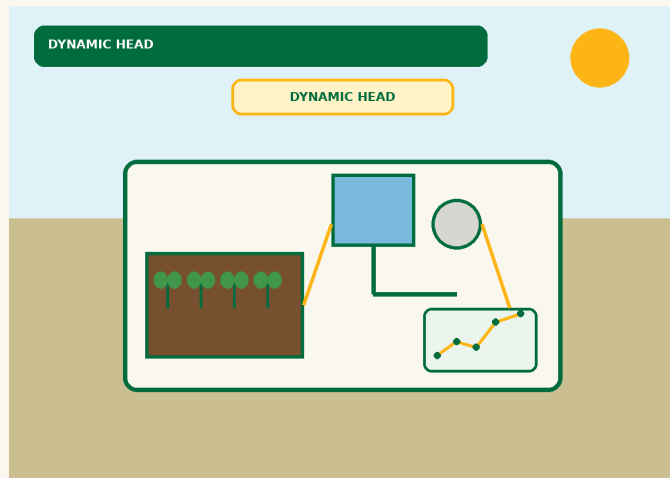
DESIGN • INSTRUMENT • MODEL • EVALUATE • OPTIMIZE

TECHNICAL CONCEPT

Static head is the elevation-related pressure potential between two points when water is not flowing.

ANALYSIS / DESIGN TASK

Calculate elevation head from a height difference.



DYNAMIC HEAD

DYNAMIC HEAD

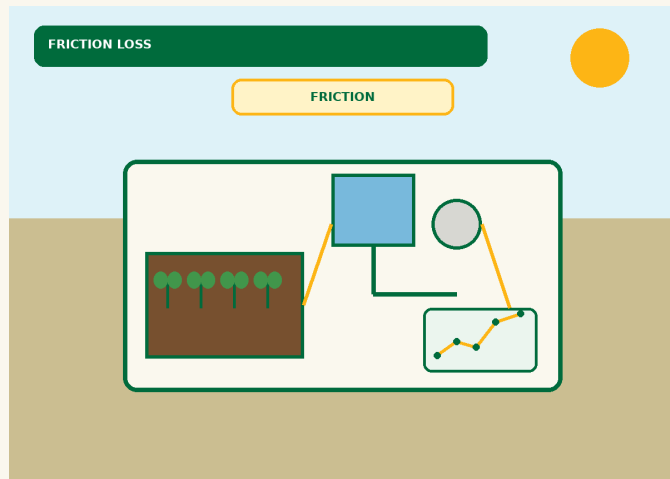
DESIGN • INSTRUMENT • MODEL • EVALUATE • OPTIMIZE

TECHNICAL CONCEPT

Dynamic head accounts for elevation, pressure requirements, and energy losses while water is flowing.

ANALYSIS / DESIGN TASK

Identify components contributing to total head.



FRICTION

FRICTION LOSS

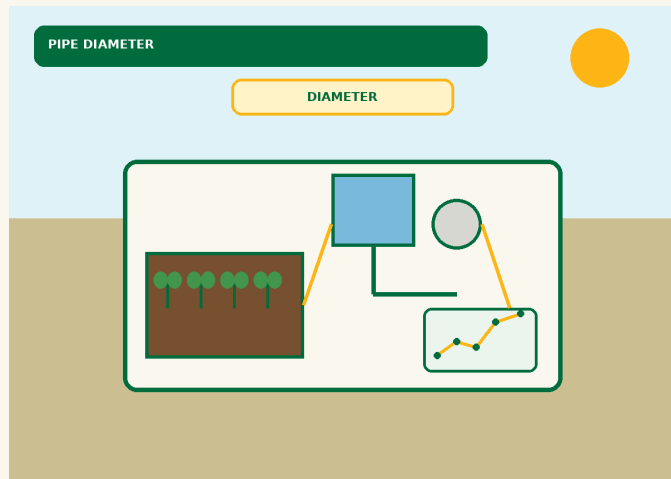
DESIGN • INSTRUMENT • MODEL • EVALUATE • OPTIMIZE

TECHNICAL CONCEPT

Friction loss is hydraulic energy lost as water moves through pipes, fittings, valves, and filters.

ANALYSIS / DESIGN TASK

Compare expected loss in two pipe sizes.



DIAMETER

PIPE DIAMETER

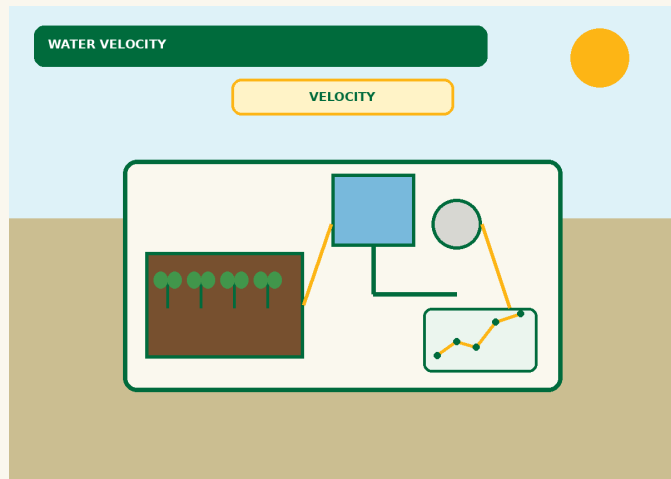
DESIGN • INSTRUMENT • MODEL • EVALUATE • OPTIMIZE

TECHNICAL CONCEPT

Pipe diameter influences velocity, friction loss, cost, and allowable system flow.

ANALYSIS / DESIGN TASK

Explain why undersized pipe increases loss.



VELOCITY

WATER VELOCITY

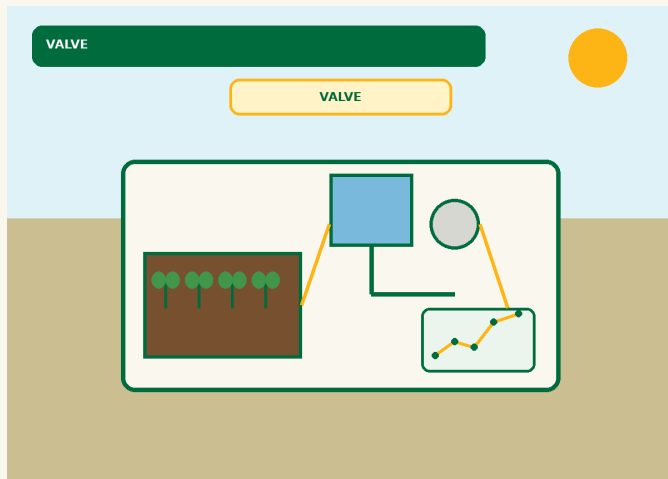
DESIGN • INSTRUMENT • MODEL • EVALUATE • OPTIMIZE

TECHNICAL CONCEPT

Water velocity is flow rate divided by pipe cross-sectional area and affects friction and hydraulic behavior.

ANALYSIS / DESIGN TASK

Calculate a simplified velocity.



VALVE

VALVE

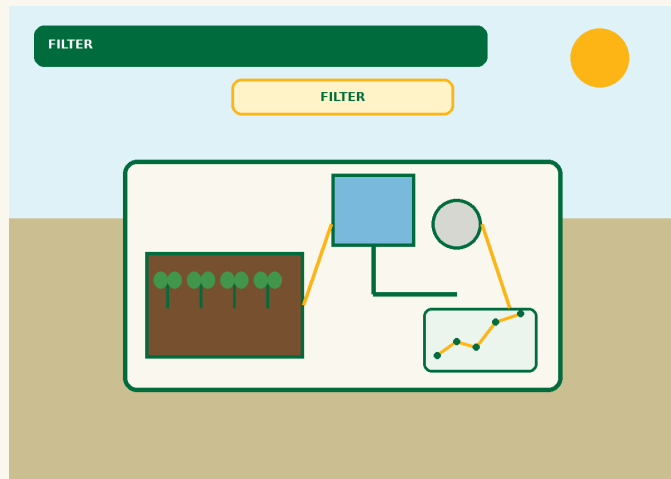
DESIGN • INSTRUMENT • MODEL • EVALUATE • OPTIMIZE

TECHNICAL CONCEPT

Valves isolate, regulate, or direct flow and are selected according to hydraulic and operational requirements.

ANALYSIS / DESIGN TASK

Identify valve functions in the system.



FILTER

FILTER

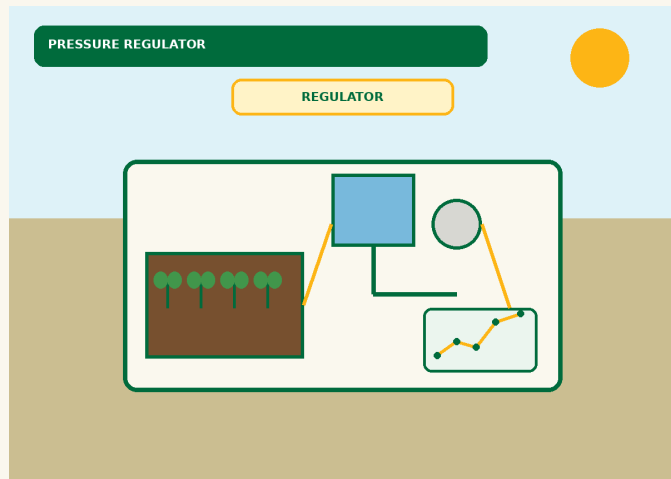
DESIGN • INSTRUMENT • MODEL • EVALUATE • OPTIMIZE

TECHNICAL CONCEPT

Filtration protects emitters and small passages from particles that can cause blockage.

ANALYSIS / DESIGN TASK

Select an appropriate maintenance check.



REGULATOR

PRESSURE REGULATOR

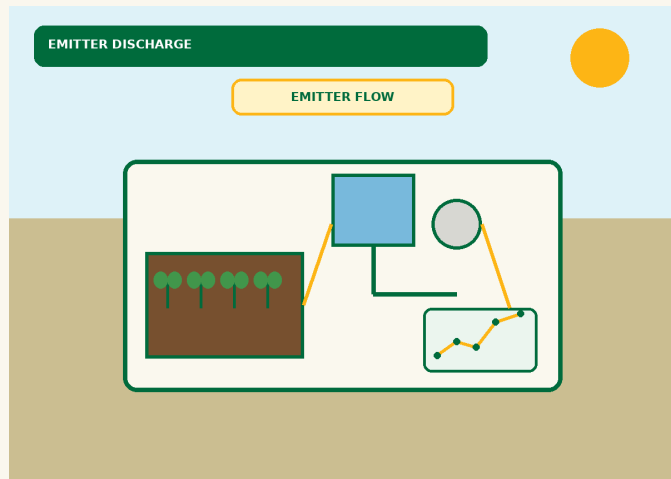
DESIGN • INSTRUMENT • MODEL • EVALUATE • OPTIMIZE

TECHNICAL CONCEPT

A pressure regulator maintains downstream pressure near a target range despite allowable upstream variation.

ANALYSIS / DESIGN TASK

Explain where regulation is needed.



EMITTER FLOW

EMITTER DISCHARGE

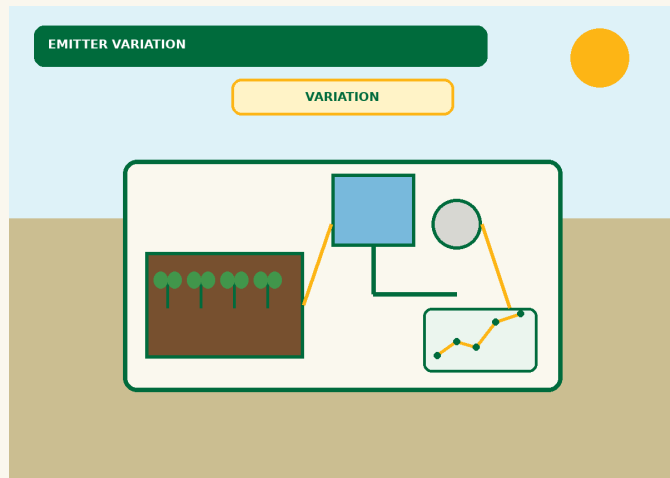
DESIGN • INSTRUMENT • MODEL • EVALUATE • OPTIMIZE

TECHNICAL CONCEPT

Emitter discharge is the volume of water released per unit time at an outlet under specified pressure conditions.

ANALYSIS / DESIGN TASK

Measure discharge using a timed collection.



VARIATION

EMITTER VARIATION

DESIGN • INSTRUMENT • MODEL • EVALUATE • OPTIMIZE

TECHNICAL CONCEPT

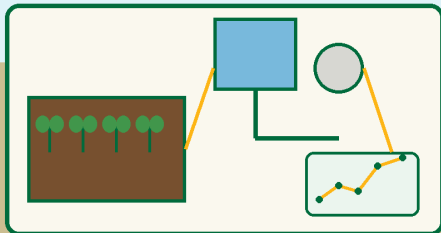
Emitter-flow variation reveals nonuniform delivery caused by pressure differences, manufacturing variation, clogging, or damage.

ANALYSIS / DESIGN TASK

Calculate maximum-to-minimum variation.

DISTRIBUTION UNIFORMITY

DU



DU

DISTRIBUTION UNIFORMITY

DESIGN • INSTRUMENT • MODEL • EVALUATE • OPTIMIZE

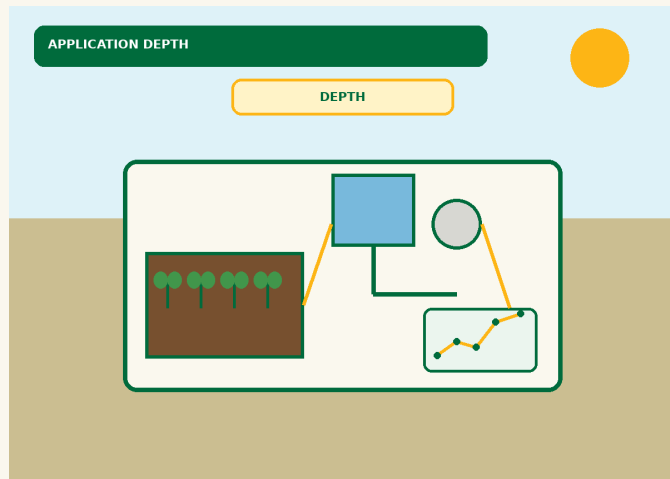
TECHNICAL CONCEPT

Distribution uniformity quantifies how evenly irrigation water is delivered across the irrigated area using a defined calculation method.

ANALYSIS / DESIGN TASK

Calculate a simplified DU from catch data.

DEFINE → MEASURE → MODEL → EVALUATE → DECIDE



DEPTH

APPLICATION DEPTH

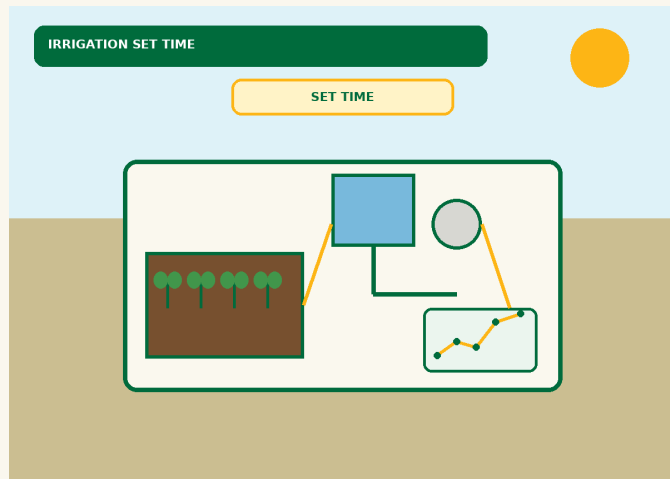
DESIGN • INSTRUMENT • MODEL • EVALUATE • OPTIMIZE

TECHNICAL CONCEPT

Application depth converts delivered water volume into an equivalent depth over the irrigated area.

ANALYSIS / DESIGN TASK

Convert litres over square metres to millimetres.



SET TIME

IRRIGATION SET TIME

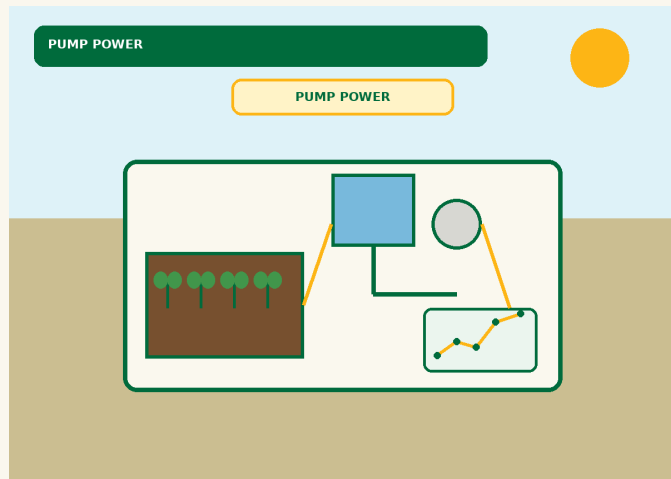
DESIGN • INSTRUMENT • MODEL • EVALUATE • OPTIMIZE

TECHNICAL CONCEPT

Set time is the operating duration needed to apply a target irrigation depth at a known application rate.

ANALYSIS / DESIGN TASK

Calculate required runtime.



PUMP POWER

PUMP POWER

DESIGN • INSTRUMENT • MODEL • EVALUATE • OPTIMIZE

TECHNICAL CONCEPT

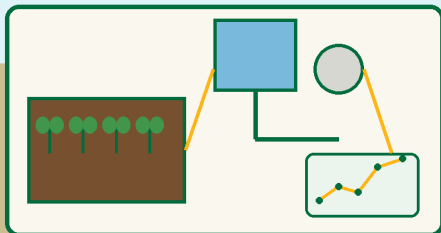
Pump power demand depends on water flow, total head, fluid properties, and pump-system efficiency.

ANALYSIS / DESIGN TASK

Interpret a simplified pump-power equation.

PUMP EFFICIENCY

PUMP EFF



PUMP EFF

PUMP EFFICIENCY

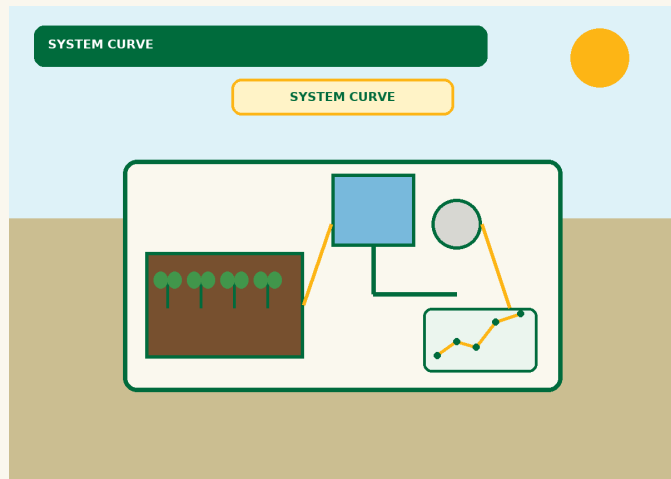
DESIGN • INSTRUMENT • MODEL • EVALUATE • OPTIMIZE

TECHNICAL CONCEPT

Pump efficiency compares useful hydraulic power delivered to water with input power to the pump system.

ANALYSIS / DESIGN TASK

Calculate efficiency from provided values.



SYSTEM CURVE

SYSTEM CURVE

DESIGN • INSTRUMENT • MODEL • EVALUATE • OPTIMIZE

TECHNICAL CONCEPT

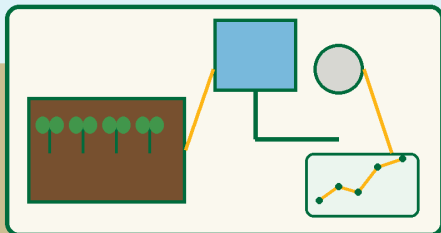
A system curve represents the head required by a piping system across different flow rates.

ANALYSIS / DESIGN TASK

Interpret a simplified system-curve plot.

IRRIGATION DESIGN REVIEW

DESIGN REVIEW



DESIGN REVIEW

IRRIGATION DESIGN REVIEW

DESIGN • INSTRUMENT • MODEL • EVALUATE • OPTIMIZE

TECHNICAL CONCEPT

Engineering review checks demand, flow, pressure, losses, uniformity, runtime, pumping requirements, safety, and maintainability.

ANALYSIS / DESIGN TASK

Evaluate a school-scale irrigation design.